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Inventor(s)

KIM; Jisung et al.

CONNECTOR AND WIRING HARNESS

Abstract

A connector (30) electrically connects a shield member to a mounting object by being fixed to an end part of a shielded cable (20) including a core wire (23), an insulation coating (24) covering an outer periphery of the core wire and the shield member (22) for covering an outer periphery of the insulation coating and fixed to the mounting object. The connector includes a metal pipe (31), through which the core wire and the insulation coating are passed and to which the shield member is connected, and a metal fixing member (32) to be fixed to an outer peripheral surface of the metal pipe and fixed to the mounting object.

Inventors:	KIM; Jisung (Mie, JP), HAMAMOTO; Naoya (Mie, JP)
Applicant:	AUTONETWORKS TECHNOLOGIES, LTD. (Mie, JP); SUMITOMO WIRING SYSTEMS, LTD. (Mie, JP); SUMITOMO ELECTRIC INDUSTRIES, LTD. (Osaka, JP)
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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to a connector and a wiring harness.

BACKGROUND

[0002] Conventionally, a wiring harness is known which is provided with a shielded cable and a connector fixed to an end part of the shielded cable (see, for example, Patent Document 1). The connector disclosed in Patent Document 1 includes a resin housing and a metal shell covering the outside of the resin housing and to be fixed to a mounting object. A shield member of the shielded cable is connected to the metal shell and electrically connected to the mounting object via the metal shell. In this way, the emission of electromagnetic waves from a core wire of the shielded cable is suppressed.

PRIOR ART DOCUMENT

Patent Document

[0003] Patent Document 1: JP 2020-155306 A

SUMMARY OF THE INVENTION

Problems to be Solved

[0004] In the above connector, the resin housing, which is a resin molded article, has been necessary. The connector is required to have an easily manufacturable structure.

[0005] The present disclosure aims to provide easily manufacturable connector and wiring harness.

Means to Solve the Problem

[0006] A connector of the present disclosure is for electrically connecting a shield member to a mounting object by being fixed to an end part of a shielded cable including a core wire, an insulation coating covering an outer periphery of the core wire and the shield member for covering an outer periphery of the insulation coating and fixed to the mounting object, and provided with a metal pipe, the core wire and the insulation coating being passed through the metal pipe, the shield member being connected to the metal pipe, and a metal fixing member to be fixed to an outer peripheral surface of the metal pipe and fixed to the mounting object.

[0007] A wiring harness of the present disclosure is provided with the connector and the shielded cable.

Effect of the Invention

[0008] According to the present disclosure, it is possible to provide easily manufacturable connector and wiring harness.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 is a perspective view of a part of a wiring harness in one embodiment.

[0010] FIG. 2 is a section of a housing of an electrical device and the wiring harness in the embodiment.

[0011] FIG. 3 is a partial exploded perspective view of the wiring harness in the embodiment.

[0012] FIG. 4 is an exploded perspective view of the connector in the embodiment.

DETAILED DESCRIPTION TO EXECUTE THE INVENTION

Description of Embodiments of Present Disclosure

[0013] First, embodiments of the present disclosure are listed and described below.

[0014] [1] The connector of the present disclosure is for electrically connecting a shield member to a mounting object by being fixed to an end part of a shielded cable including a core wire, an insulation coating covering an outer periphery of the core wire and the shield member for covering an outer periphery of the insulation coating and fixed to the mounting object, and provided with a metal pipe, the core wire and the insulation coating being passed through the metal pipe, the shield member being connected to the metal pipe, and a metal fixing member to be fixed to an outer peripheral surface of the metal pipe and fixed to the mounting object.

[0015] According to this configuration, the shield member is connected to the mounting object via the metal pipe and the metal fixing member while the core wire is covered by the metal pipe. Thus, the emission of electromagnetic waves from the core wire can be suppressed. Here, since the metal pipe and the metal fixing member are separate bodies and fixed to each other after manufacturing, each of these can be easily obtained. The connector of the embodiment does not require a resin molded article such as a resin housing provided in conventional connectors and, consequently, a mold for molding this resin molded article is also not required. Therefore, the connector of the embodiment can be easily manufactured.

[0016] [2] Preferably, in [1] described above, the metal fixing member is made of a plate material.

[0017] According to this configuration, the metal fixing member can be more easily manufactured as compared to a cast.

[0018] [3] Preferably, in [1] or [2] described above, the metal fixing member includes a pipe fitting hole, and the metal pipe includes a first pipe expansion portion to be coupled to an inner surface of the pipe fitting hole.

[0019] According to this configuration, since the metal pipe can be coupled to the inner surface of the pipe fitting hole by applying a pipe expansion processing to the metal pipe, a fixing operation is easier as compared to welding.

[0020] [4] Preferably, in any one of [1] to [3] described above, a first rubber ring is provided which is externally fit to the metal pipe.

[0021] According to this configuration, by interposing the first rubber ring between the outer peripheral surface of the metal pipe and the mounting object, this part can be waterproofed.

[0022] [5] Preferably, in [4] described above, the metal pipe includes a first pipe contraction portion, the first rubber ring being externally fit to the first pipe contraction portion.

[0023] According to this configuration, a position shift of the first rubber ring along a length direction of the metal pipe is suppressed.

[0024] [6] Preferably, in any one of [1] to [5] described above, a second rubber ring is provided which is fit into the metal pipe.

[0025] According to this configuration, by interposing the second rubber ring between the inner peripheral surface of the metal pipe and the outer peripheral surface of the insulation coating, this part can be waterproofed.

[0026] [7] Preferably, in [6] described above, the metal pipe includes a second pipe expansion portion, the second rubber ring being fit into the second pipe contraction portion.

[0027] According to this configuration, a position shift of the second rubber ring along the length direction of the metal pipe is suppressed.

[0028] [8] Preferably, in any one of [1] to [7] described above, the metal pipe includes a second pipe contraction portion to be pressed into contact with the insulation coating.

[0029] According to this configuration, the insulation coating and the connector can be fixed by the second pipe contraction portion without using a separate member.

[0030] [9] Preferably, in any one of [1] to [8] described above, the shielded cable is a multi-core collective shielded cable including a plurality of the core wires and a plurality of the insulation coatings, the shield member collectively covering the plurality of insulation coatings, a plurality of

the metal pipes are provided, a set of the core wire and the insulation coating being passed through each metal pipe, the metal fixing member is so fixed to at least one of the plurality of metal pipes that axes of the plurality of metal pipes are parallel to each other, and a connecting member is provided to fill up gaps between outer peripheral surfaces of the plurality of metal pipes, the shield member being connected to the connecting member.

[0031] According to this configuration, the shield member in the multi-core collective shielded cable can be satisfactorily electrically connected to the metal pipes by the connecting member. That is, in a configuration in which the connecting member is not provided, a contact area is small since the shield member is not in contact with or difficult to contact the plurality of metal pipes in the gaps between the outer peripheral surfaces of the plurality of metal pipes. However, this can be avoided in the above configuration.

[0032] [10] Preferably, in [7] described above, the shielded cable is a multi-core collective shielded cable including a plurality of the core wires and a plurality of the insulation coatings, the shield member collectively covering the plurality of insulation coatings, a plurality of the metal pipes are provided, a set of the core wire and the insulation coating being passed through each metal pipe, the metal fixing member is so fixed to at least one of the plurality of metal pipes that axes of the plurality of metal pipes are parallel to each other, and a connecting member is provided to fill up gaps between outer peripheral surfaces of the second pipe expansion portions in the plurality of metal pipes, the shield member being connected to the connecting member.

[0033] According to this configuration, in addition to functions and effects of [9], the material cost of the connecting member provided to fill up the gaps can be reduced since the gaps between the outer peripheral surfaces of the second pipe expansion portions are smaller than gaps between the outer peripheral surfaces of other parts, to which the pipe expansion processing is not applied.

[0034] [11] The wiring harness of the present disclosure is provided with the connector of any one of [1] to described above and the shielded cable.

[0035] According to this configuration, functions and effects similar to those of the connector are obtained.

[0036] [12] The wiring harness is provided with the connector of [9] or described above, the shielded cable, and a crimping member for fixing the shield member to the connecting member by being crimped.

[0037] According to this configuration, in addition to functions and effects of [11], the shield member can be easily connected to the connecting member by the crimping member.

DETAILS OF EMBODIMENTS OF PRESENT DISCLOSURE

[0038] A specific example of a wiring harness of the present disclosure is described below with reference to the drawings. For the convenience of description, some components may be shown in an exaggerated or simplified manner in each drawing. Further, a dimension ratio of each part may be different in each drawing. “Parallel”, “orthogonal” and “true circular” in this specification mean not only strictly parallel, orthogonal and true circular, but also substantially parallel, orthogonal and true circular within a range in which functions and effects in this embodiment are achieved. Note that the present invention is not limited to these illustrations, but is represented by claims and intended to include all changes in the scope of claims and in the meaning and scope of equivalents.

(Configuration of Wiring Harness **10**)

[0039] As shown in FIG. **1**, a wiring harness **10** is provided with a multi-core collective shielded cable **20** serving as a shielded cable and a connector **30**.

(Configuration of Multi-Core Collective Shielded Cable **20**)

[0040] As shown in FIGS. **1** to **3**, the multi-core collective shielded cable **20** includes wires **21** and a shield member **22**. The multi-core collective shielded cable **20** of this embodiment includes three wires **21**.

[0041] The wire **21** includes a core wire **23** and an insulation coating **24** covering the outer periphery of the core wire **23**. The core wire **23** of this embodiment has a circular cross-sectional

shape. Further, the insulation coating **24** has an annular cross-sectional shape. The core wire **23** may be made of an electrically conductive material such as metal.

[0042] The shield member **22** covers the outer periphery of the insulation coating **24**. The shield member **22** of this embodiment collectively covers a plurality of the insulation coatings **24**. The shield member **22** is, for example, a braided wire formed by braiding electrically conductive strands of copper alloy, aluminum alloy or the like into a tube. Note that the shield member **22** is schematically shown in each figure. An end part of the wire **21** projects outward from an end part of the shield member **22**.

(Configuration of Connector **30**)

[0043] As shown in FIG. **2**, the connector **30** is fixed to an end part of the multi-core collective shielded cable **20** and a housing **50** of an electrical device serving as a mounting object.

[0044] As shown in FIGS. **1** to **4**, the connector **30** is provided with metal pipes **31** and a metal fixing member **32** to be fixed to the outer peripheral surfaces of the metal pipes **31**. The connector **30** of this embodiment is provided with three metal pipes **31** corresponding to the three wires **21** of the multi-core collective shielded cable **20**. That is, the connector **30** is provided with the three metal pipes **31** and one wire **21** is passed through each metal pipe **31**. In other words, a set of the core wire **23** and the insulation coating **24** is passed through each metal pipe **31**.

(Configuration of Metal Fixing Member **32**)

[0045] The metal fixing member **32** is made of a plate material. In particular, the metal fixing member **32** is configured, such as by stamping the metal plate material.

[0046] As shown in FIG. **4**, the metal fixing member **32** includes a body portion **32a** having a rectangular shape when viewed from a plate thickness direction and fixing portions **32b** projecting in a short-side direction from both end sides in a longitudinal direction of the body portion **32a**. The metal fixing member **32** includes pipe fixing holes **32c** penetrating in the plate thickness direction. Three pipe fixing holes **32c** are formed in the body portion **32a**. The three pipe fixing holes **32c** are provided side by side at equal intervals along the longitudinal direction of the body portion **32a**. Note that an unevenness processing such as knurling is preferably applied to the inner surface of the pipe fixing hole **32c**. Further, the metal fixing member **32** includes connector mounting holes **32d** penetrating in the plate thickness direction. The connector mounting hole **32d** is formed in each of the two fixing portions **32b**.

(Configuration of Metal Pipe **31**)

[0047] The metal pipe **31** is made of a pipe material. In Particular, the metal pipe **31** is configured, such as by applying a pipe expansion processing of expanding a diameter or a pipe contraction processing of reducing the diameter to the pipe material having a constant diameter and made of metal.

[0048] As shown in FIG. **2**, the metal pipe **31** includes a first pipe expansion portion **31a** to be coupled to the inner surface of the pipe fixing hole **32c**. The first pipe expansion portion **31a** is a part expanded to have a larger diameter than adjacent parts in a length direction in the metal pipe **31**. The first pipe expansion portion **31a** is provided in an intermediate part of the metal pipe **31**. The metal pipe **31** is fixed to the metal fixing member **32** by pressing the first pipe expansion portion **31a** into contact with and coupled to the inner surface of the pipe fixing hole **32c** when the first pipe expansion portion **31a** is formed by the pipe expansion processing. In other words, the metal fixing member **32** is fixed to the metal pipe **31** by expanding the metal pipe **31** in the pipe fixing hole **32c** to form the first pipe expansion portion **31a**.

[0049] As shown in FIGS. **3** and **4**, the metal fixing member **32** is so fixed to a plurality of the metal pipes **31** that axes Z of the plurality of metal pipes **31** are parallel to each other. Note that FIG. **4** is not an exploded perspective view of a state before the metal pipes **31** and the metal fixing member **32** are fixed, but shows the metal pipes **31** including the first pipe expansion portions **31a** in a state where the metal pipes **31** and the metal fixing member **32** are disassembled again after being once fixed.

[0050] Further, the metal pipe **31** includes a first pipe contraction portion **31b**. The first pipe contraction portion **31b** is a part contracted to have a smaller diameter than adjacent parts in the length direction in the metal pipe **31**. The first pipe contraction portion **31b** is provided on a side closer to the tip of the metal pipe **31** than the first pipe expansion portion **31a**. In particular, the metal pipe **31** includes a first end part **31c** arranged on a tip side of the wiring harness **10** and a second end part **31d** arranged on an intermediate side of the wiring harness **10**. The first pipe contraction portion **31b** is provided on a side closer to the first end part **31c** than the first pipe expansion portion **31a**.

[0051] The connector **30** is provided with first rubber rings **33** to be externally fit to the metal pipes **31**. The first rubber ring **33** is externally fit to the first pipe contraction portion **31b**.

[0052] Further, the metal pipe **31** includes a second pipe expansion portion **31e**. The second pipe expansion portion **31e** is a part expanded to have a larger diameter than adjacent parts in the length direction in the metal pipe **31**. The second pipe expansion portion **31e** is provided on a side closer to the second end part **31d** than the first pipe expansion portion **31a** while being spaced apart from the first pipe expansion portion **31a**. Note that a basis portion **31f** maintained to have a diameter of the pipe material constituting the metal pipe **31** without the pipe expansion processing or the pipe contraction processing being applied thereto is provided between the first and second pipe expansion portions **31a**, **31b**.

[0053] The connector **30** is provided with second rubber rings **34** to be fit into the metal pipes **31**. The second rubber ring **34** is fit into the second pipe expansion portion **31e**.

[0054] The wire **21** is inserted through the metal pipe **31**. With the wire **21** inserted through the metal pipe **31**, the second rubber ring **34** is interposed in a compressed state between the outer peripheral surface of the insulation coating **24** and the inner peripheral surface of the metal pipe **31**.

[0055] Further, the metal pipe **31** includes a second pipe contraction portion **31g**. The second pipe contraction portion **31g** is a part contracted to have a smaller diameter than adjacent parts in the length direction in the metal pipe **31**. The second pipe contraction portion **31g** is provided on the second end part **31d**. The metal pipe **31** is fixed to the wire **21** by pressing the second pipe contraction portion **31g** into contact with the insulation coating **24**.

[0056] As shown in FIGS. **3** and **4**, the connector **30** is provided with a connecting member **35**. As shown in FIG. **4**, the connecting member **35** includes three through holes **35a**, into which the metal pipes **31** are respectively fit. The through hole **35a** is formed in such a size that the second pipe expansion portion **31e** of the metal pipe **31** is fit thereinto. The connecting member **35** is provided to fill up gaps **36** between the outer peripheral surfaces **31h** of the three metal pipes **31** while covering the three metal pipes **31**. In particular, the connecting member **35** is provided to fill up the gaps **36** between the outer peripheral surfaces **31h** of the second pipe expansion portions **31e** in the three metal pipes **31** while covering the second pipe expansion portions **31e** of the three metal pipes **31**. That is, the connecting member **35** fills up the gaps **36** formed between the outer peripheral surfaces **31h** of the metal pipes **31** by coupling parts **35b** between the through holes **35a**.

[0057] As shown in FIG. **2**, the shield member **22** is connected to the outer peripheral surface of the connecting member **35**. In particular, the wiring harness **10** is provided with a crimping member **37**. The shield member **22** is arranged to cover the outer peripheral surface of the connecting member **35** and, further, connected to the connecting member **35** by crimping the crimping member **37** to the outside of the shield member **22**. That is, the connecting member **35** contacts the shield member **22** over an entire periphery while serving as a base when the crimping member **37** is crimped. The crimping member **37** fixes the shield member **22** to the connecting member **35** by being crimped. Note that the crimping member **37** of this embodiment is a crimp ring made of metal and formed into an annular shape. The shield member **22** is electrically connected to the metal pipes **31** via the connecting member **35**.

[0058] Further, an end part of the core wire **23** is exposed to outside from the first end part **31c** of the metal pipe **31** and exposed from the insulation coating **24**. A terminal **38** is crimped and fixed to

the end part of the core wire **23**.

(Configuration of Housing **50** of Electrical Device and Fixing Structure of Connector **30**)

[0059] As shown in FIG. **2**, the housing **50** of the electrical device serving as the mounting object is provided with mounting holes **50a** and fastening portions **50b**. The housing **50** of the electrical device is made of metal. The connector **30** is so fixed to the housing **50** of the electrical device that the terminals **38** and the sides of the first end parts **31c** of the metal pipes **31** are inserted in the mounting holes **50a**.

[0060] With the sides of the first end parts **31c** of the metal pipes **31** inserted in the mounting holes **50a**, the first rubber rings **33** are interposed in a compressed state between the outer peripheral surfaces of the metal pipes **31** and the inner peripheral surfaces of the mounting holes **50a**. The connector **30** is fixed to the housing **50** of the electrical device by bolts **39** threadably engaged with the fastening portions **50b** through the connector mounting holes **32d**. In this way, the shield member **22** is electrically connected to the housing **50** of the electrical device via the connector **30**. With the connector **30** fixed to the housing **50** of the electrical device, unillustrated terminals of the electrical device are connected to the terminals **38**.

[0061] Next, functions of the wiring harness **10** configured as described above are described.

[0062] In the multi-core collective shielded cable **20**, the insulation coatings **24** of the plurality of wires **21** are collectively covered by the shield member **22**. Thus, the emission of electromagnetic waves from the cores **23** in parts covered by the shield member **22** is suppressed. Further, the shield member **22** is electrically connected to the housing **50** of the electrical device by fixing the connector **30** to the housing **50** of the electrical device. In the connector **30**, the wires **21** are covered by the metal pipes **31**. Thus, the emission of electromagnetic waves from parts of the core wires **23** in the connector **30** is suppressed.

[0063] Next, effects of the above embodiment are described below.

[0064] (1) In the connector **30**, the core wires **23** are covered by the metal pipes **31**. Further, the shield member **22** is connected to the housing **50** of the electrical device via the metal pipes **31** and the metal fixing member **32**. Thus, the emission of electromagnetic waves from the core wires **23** can be suppressed. Since the metal pipes **31** and the metal fixing member **32** are separate bodies and fixed to each other after manufacturing, each of these can be easily obtained. The connector **30** of the embodiment does not require a resin molded article such as a resin housing provided in conventional connectors and, consequently, a mold for molding this resin molded article is also not required. Therefore, the connector **30** of the embodiment can be easily manufactured.

[0065] (2) The metal fixing member **32** is made of the plate material. Thus, the metal fixing member **32** can be more easily manufactured, for example, as compared to a cast.

[0066] (3) The metal pipe **31** includes the first pipe expansion processing **31a** to be coupled to the inner surface of the pipe fixing hole **32c**, and can be coupled to the inner surface of the pipe fixing hole **32c** by applying the pipe expansion processing to the metal pipe **31**. Thus, a fixing operation is easier, for example, as compared to welding.

[0067] (4) The first rubber rings **33** are provided which are externally fit to the metal pipes **31**. Thus, by interposing the first rubber ring **33** between the outer peripheral surface of the metal pipe **31** and the inner peripheral surface of the mounting hole **50a** in the housing **50** of the electrical device, this part can be waterproofed. Since the first rubber rings **33** can have a simple shape without needing any special shape and can be assembled by being externally fit to the metal pipes **31**, the connector **30** can be easily manufactured.

[0068] (5) The metal pipe **31** includes the first pipe contraction portion **31b**, to which the first rubber ring **33** is externally fit. Thus, a position shift of the first rubber ring **33** along the length direction of the metal pipe **31** is suppressed.

[0069] (6) The second rubber ring **34** is provided which is fit into the metal pipe **31**. Thus, by interposing the second rubber ring **34** between the inner peripheral surface of the metal pipe **31** and the outer peripheral surface of the insulation coating **24**, this part can be waterproofed. Since the

second rubber rings **34** can have a simple shape without needing any special shape and can be assembled by being fit into the metal pipes **31**, the connector **30** can be easily manufactured. [0070] (7) The metal pipe **31** includes the second pipe expansion portion **31e**, into which the second rubber ring **34** is fit. Thus, a position shift of the second rubber ring **34** along the length direction of the metal pipe **31** is suppressed.

[0071] (8) The metal pipe **31** includes the second pipe contraction portion **31g** to be pressed into contact with the insulation coating **24**. Thus, the insulation coating **24** and the connector **30** can be fixed by the second pipe contraction portion **31g** without using a separate member.

[0072] (9) The connecting member **35** is provided which fills up the gaps **36** between the outer peripheral surfaces **31h** of the plurality of metal pipes **31** and to which the shield member **22** is connected. Thus, the shield member **22** in the multi-core collective shielded cable **20** can be satisfactorily electrically connected to the metal pipes **31**. That is, in a configuration in which the connecting member **35** is not provided, a contact area is small since the shield member **22** is not in contact with or difficult to contact the plurality of metal pipes **31** in the gaps **36** between the outer peripheral surfaces **31h** of the plurality of metal pipes **31**. However, this can be avoided in the above configuration. Therefore, the shield member **22** can be satisfactorily connected to the metal pipes **31**.

[0073] (10) The connecting member **35** is provided to fill up the gaps **36** between the outer peripheral surfaces **31h** of the second pipe expansion portions **31e** in the metal pipes **31**. Since the gaps **36** between the outer peripheral surfaces **31h** of the second pipe expansion portions **31e** are smaller than gaps between the outer peripheral surfaces of other parts, to which the pipe expansion processing is not applied, the material cost of the connecting member **35** provided to fill up the gaps **36** can be reduced.

[0074] (11) The shield member **22** can be easily connected to the connecting member **35** by the crimping member **37**.

[0075] This embodiment can be modified and carried out as follows. This embodiment and the following modifications can be carried out in combination with technically contradicting each other. [0076] Although the metal fixing member **32** is made of the plate material in the above embodiment, there is no limitation to this and the metal fixing member **32** may be a cast or the like formed using a mold. Further, although the metal fixing member **32** is configured such as by stamping the metal plate material, another processing such as bending may be applied while the metal fixing member **32** is made of the plate material. [0077] Although the metal pipe **31** includes the first pipe expansion portion **31a** to be coupled to the inner surface of the pipe fixing hole **32c** in the above embodiment, there is no limitation to this and the first pipe expansion portion **31a** may not be provided. For example, a metal pipe and a metal fixing member may be fixed by welding. [0078] Although the connector **30** is provided with the first rubber rings **33** to be externally fit to the metal pipes **31** in the above embodiment, there is no limitation to this and the first rubber rings **33** may not be provided. For example, a sealing member may be fixed to the metal fixing member **32** by adhesion or the like instead of using the first rubber rings **33**. [0079] Although the metal pipe **31** includes the first pipe contraction portion **31b**, to which the first rubber ring **33** is externally fit, in the above embodiment, there is no limitation to this and the first pipe contraction portion **31b** may not be provided. [0080] Although the connector **30** is provided with the second rubber rings **34** to be fit into the metal pipes **31** in the above embodiment, there is no limitation to this and the second rubber rings **34** may not be provided. For example, a sealing member may be fixed to the second end parts **31d** of the metal pipes **31** by adhesion or the like instead of using the second rubber rings **34**. [0081] Although the metal pipe **31** includes the second pipe expansion portion **31e**, into which the second rubber ring **34** is fit, in the above embodiment, there is no limitation to this and the second pipe expansion portion **31e** may not be provided. Further, the first pipe expansion portion **31a** may be used as a second pipe expansion portion without independently forming the second pipe expansion portion **31e** and the second rubber ring **34** may be fit into the first pipe

expansion portion **31e** constituting the second pipe expansion portion. [0082] Although the metal pipe **31** includes the second pipe contraction portion **31g** to be pressed into contact with the insulation coating **24** in the above embodiment, there is no limitation to this and the second pipe contraction portion **31g** may not be provided. For example, the insulation coating **24** and, consequently, the wire **21** may be fixed to the metal pipe **31** by another member. [0083] Although the connector **30** includes the connecting member **35** in the above embodiment, there is no limitation to this and the connecting member **35** may not be provided. Particularly, since the shielded cable is the multi-core collective shielded cable **20** in the above embodiment, the connecting member **35** is provided to satisfactorily electrically connect the shield member **22** to the metal pipes **31**. However, the connecting member **35** is unnecessary in a configuration in which an independent shield member is provided for each wire **21**. That is, in a configuration in which the shielded cable is each core shielded cable and the independent shield member is provided for each wire **21**, each shield member may be directly connected to each metal pipe **31**. [0084] Although the connecting member **35** is provided to fill up the gaps **36** between the outer peripheral surfaces **31h** of the second pipe expansion portions **31e** in the above embodiment, there is no limitation to this and the connecting member **35** may be provided to fill up gaps between the outer peripheral surfaces of parts of the metal pipes **31** other than the second pipe expansion portions **31e**. [0085] Although the crimping member **37** is provided in the above embodiment, there is no limitation to this. For example, the shield member **22** may be fixed to the connecting member **35** by welding without providing the crimping member **37**. [0086] Although the multi-core collective shielded cable **20** includes the three wires **21** and the connector **30** is provided with the three metal pipes **31** in the above embodiment, there is no limitation to this and the number of the wires **21** and the number of the metal pipes **31** may be changed to another number. [0087] The metal pipes **31** and the metal fixing member **32** may be configured as separate bodies and the metal pipes **31** and the metal fixing member **32** may be fixed by crimping the outer peripheral surfaces of the metal pipes **31** to the inner surfaces of the pipe fixing holes **32c** of the metal fixing members **32**. [0088] The connecting member **35** may be an electrically conductive member for electrically connecting the metal pipes **31** and the shield member **22**. The connecting member **35** may have an inner peripheral surface to be brought into direct contact with the outer peripheral surfaces of the metal pipes **31**. The connecting member **35** may have an outer peripheral surface to be brought into direct contact with the inner peripheral surface of the shield member **22**. [0089] As shown in FIG. 2, the first pipe contraction portion **31b** of the metal pipe **31** can form a step for suppressing a position shift of the first rubber ring **33** along the length direction of the metal pipe **31** toward the first end part **31c**. The first rubber ring **33** externally fit to the first pipe contraction portion **31b** may project more radially outward of the metal pipe **31** than the first pipe expansion portion **31a** and the first end part **31c** of the metal pipe **31**. [0090] A connector (**30**) according to several embodiments of the present disclosure may be configured to be used together with an electrical device including a metal housing (**50**) including a mounting hole (**50a**) configured to receive a part of the connector (**30**) and an electrically conductive connector receiving surface to be brought into contact with another part of the connector (**30**), and may be provided with: a metal fixing member (**32**) including a pipe fixing hole (**32c**) to be aligned with the mounting hole (**50a**) of the metal housing (**50**), the metal fixing member (**32**) being fixed to the electrically conductive connector receiving surface of the metal housing (**50**); a shielded cable (**20**) including an electrically conductive core wire (**23**) having a tip part, an insulation coating (**24**) covering the core wire (**23**) in such a manner as to expose the tip part of the core wire (**23**) and an electrically conductive shield member (**22**) for exposing the tip part of the core wire (**23**) and covering the insulation coating (**24**) in such a manner as to expose one end part of the insulation coating (**24**); and a metal pipe (**31**) for allowing the passage of the shielded cable (**20**) therethrough, the metal pipe (**31**) including a first length part, which can be the second pipe contraction portion (**31g**) of the embodiment, a second length part, which can be the second pipe expansion portion (**31e**) of the embodiment, a third length part, which can be the first

pipe expansion portion (31a) of the embodiment, and a fourth length part, which can be the first pipe contraction portion (31b) of the embodiment, the first length part of the metal pipe (31) may be configured to contact or face the outer peripheral surface of the insulation coating (24) of the shielded cable (20) and be surrounded by the shield member (22) of the shielded cable (20), the second length part of the metal pipe (31) may be configured to be electrically connected to the inner peripheral surface of the shield member (22), the third length part of the metal pipe (31) may be configured to directly contact and be electrically connected to the pipe fitting hole (32c) of the metal fixing member (32), and the fourth length part of the metal pipe (31) may be configured to be received into the mounting hole (50a) of the metal housing (50). [0091] In several embodiments of the present disclosure, the third length part of the metal pipe (31) may be directly in contact with or fixed to the pipe fitting hole (32c) of the metal fixing member (32) so that the metal pipe (31) does not move in a length direction with respect to the metal fixing member (32). [0092] In several embodiments of the present disclosure, the metal pipe (31) may be a metal one-piece product having openings on both ends.

LIST OF REFERENCE NUMERALS

[0093] **10** wiring harness [0094] **20** multi-core collective shielded cable (shielded cable) [0095] **21** wire [0096] **22** shield member [0097] **23** core wire [0098] **24** insulation coating [0099] **30** connector [0100] **31** metal pipe [0101] **31a** first pipe expansion portion [0102] **31b** first pipe contraction portion [0103] **31c** first end part [0104] **31d** second end part [0105] **31e** second pipe expansion portion [0106] **31f** basis portion [0107] **31g** second pipe contraction portion [0108] **31h** outer peripheral surface [0109] **32** metal fixing member [0110] **32a** body portion [0111] **32b** fixing portion [0112] **32c** pipe fitting hole [0113] **32d** connector mounting hole [0114] **33** first rubber ring [0115] **34** second rubber ring [0116] **35** connecting member [0117] **35a** through hole [0118] **35b** coupling part [0119] **36** gap [0120] **37** crimping member [0121] **38** terminal [0122] **39** bolt [0123] **50** housing of electrical device (mounting object) [0124] **50a** mounting hole [0125] **50b** fastening portion [0126] Z axis

Claims

1. A connector for electrically connecting a shield member to a mounting object by being fixed to an end part of a shielded cable including a core wire, an insulation coating covering an outer periphery of the core wire and the shield member for covering an outer periphery of the insulation coating and fixed to the mounting object, comprising: a metal pipe, the core wire and the insulation coating being passed through the metal pipe, the shield member being connected to the metal pipe; and a metal fixing member to be fixed to an outer peripheral surface of the metal pipe and fixed to the mounting object.
2. The connector of claim 1, wherein the metal fixing member is made of a plate material.
3. The connector of claim 1, wherein: the metal fixing member includes a pipe fitting hole, and the metal pipe includes a first pipe expansion portion to be coupled to an inner surface of the pipe fitting hole.
4. The connector of claim 1, comprising a first rubber ring to be externally fit to the metal pipe.
5. The connector of claim 4, wherein the metal pipe includes a first pipe contraction portion, the first rubber ring being externally fit to the first pipe contraction portion.
6. The connector of claim 1, comprising a second rubber ring to be fit into the metal pipe.
7. The connector of claim 6, wherein the metal pipe includes a second pipe expansion portion, the second rubber ring being fit into the second pipe contraction portion.
8. The connector of claim 1, wherein the metal pipe includes a second pipe contraction portion to be pressed into contact with the insulation coating.
9. The connector of claim 1, wherein: the shielded cable is a multi-core collective shielded cable including a plurality of the core wires and a plurality of the insulation coatings, the shield member

collectively covering the plurality of insulation coatings, a plurality of the metal pipes are provided, a set of the core wire and the insulation coating being passed through each metal pipe, the metal fixing member is so fixed to at least one of the plurality of metal pipes that axes of the plurality of metal pipes are parallel to each other, and a connecting member is provided to fill up gaps between outer peripheral surfaces of the plurality of metal pipes, the shield member being connected to the connecting member.

10. The connector of claim 7, wherein: the shielded cable is a multi-core collective shielded cable including a plurality of the core wires and a plurality of the insulation coatings, the shield member collectively covering the plurality of insulation coatings, a plurality of the metal pipes are provided, a set of the core wire and the insulation coating being passed through each metal pipe, the metal fixing member is so fixed to at least one of the plurality of metal pipes that axes of the plurality of metal pipes are parallel to each other, and a connecting member is provided to fill up gaps between outer peripheral surfaces of the second pipe expansion portions in the plurality of metal pipes, the shield member being connected to the connecting member.

11. A wiring harness, comprising: the connector of claim 1; and the shielded cable.

12. A wiring harness, comprising: the connector of claim 9; the shielded cable; and a crimping member for fixing the shield member to the connecting member by being crimped.
